

Ontology Engineering

Assumptions

- formal taxonomy of road signs as an IS-A class hierarchy, written in OWL2
- conformant to Part I, Annex 1, of Vienna Convention (document “Conv_road_signs_2006v_EN.pdf”)
- a first set of competency questions that will guide your ontology development (covering the taxonomy, properties, pragmatics, and the particular dataset)

The Dataset

- Mapillary Traffic Sign Dataset (<https://www.mapillary.com/dataset/trafficsign>)
 - 100,000 high-resolution images (52,000 fully annotated, 48,000 partially annotated)
 - Over 300 traffic sign classes
- On the Virtual Campus: Turtle file of the RDF graph covering the metadata of the fully annotated images (original annotations in JSON).


```
@prefix mtsd: <https://example.org/mtsd/> .
@prefix mtsp: <https://example.org/mtsd/property/> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
```

```
<https://example.org/mtsd/image/--7fWq6WjZM8L1eUSuvOEA>
  a mtsd:Image ;
  mtsd:sourceFile "--7fWq6WjZM8L1eUSuvOEA.json" ;
  mtsd:width "4160"^^xsd:integer ;
  mtsd:height "3120"^^xsd:integer ;
  mtsd:isPano "false"^^xsd:boolean ;
  mtsd:hasObject <https://example.org/mtsd/object/--7fWq6WjZM8L1eUSuvOEA/iw8oed2vtpa46rfvmki2wk>,
                <https://example.org/mtsd/object/--7fWq6WjZM8L1eUSuvOEA/pyiahdn8h67q6s4lt90bn3>,
                <https://example.org/mtsd/object/--7fWq6WjZM8L1eUSuvOEA/7p50jhf0w90ybtkau7w154>,
                <https://example.org/mtsd/object/--7fWq6WjZM8L1eUSuvOEA/8rynh9ttz4tubjvhnb1a2o> .
```

```
<https://example.org/mtsd/object/--7fWq6WjZM8L1eUSuvOEA/iw8oed2vtpa46rfvmki2wk>
  a mtsd:TrafficSignObject ;
  mtsd:key "iw8oed2vtpa46rfvmki2wk" ;
  rdfs:label "regulatory--keep-right--g1" ;
  mtsd:xmin "2187.65625"^^xsd:decimal ;
  mtsd:ymin "1669.6875"^^xsd:decimal ;
  mtsd:xmax "2225.234375"^^xsd:decimal ;
  mtsd:ymax "1707.7734375"^^xsd:decimal ;
  mtsp:barrier "false"^^xsd:boolean ;
  mtsp:occluded "false"^^xsd:boolean ;
  mtsp:out-of-frame "false"^^xsd:boolean ;
  mtsp:exterior "false"^^xsd:boolean ;
  mtsp:ambiguous "false"^^xsd:boolean ;
  mtsp:included "false"^^xsd:boolean ;
  mtsp:direction-or-information "false"^^xsd:boolean ;
  mtsp:highway "false"^^xsd:boolean ;
  mtsp:dummy "false"^^xsd:boolean .
```

MTSD Class Labels

- regulatory--bicycles-only--g1
- regulatory--bicycles-only--g2
- regulatory--bicycles-only--g3
- regulatory--buses-only--g1
- regulatory--detour-left--g1
- regulatory--do-not-block-intersection--g1
- regulatory--do-not-stop-on-tracks--g1
- regulatory--dual-lanes-go-straight-on-left--g1
- regulatory--dual-lanes-go-straight-on-right--g1
- regulatory--dual-lanes-turn-left-no-u-turn--g1
- regulatory--dual-lanes-turn-left-or-straight--g1
- regulatory--dual-lanes-turn-right-or-straight--g1
- regulatory--dual-path-bicycles-and-pedestrians--g1
- regulatory--dual-path-pedestrians-and-bicycles--g1
- regulatory--end-of-bicycles-only--g1
- regulatory--end-of-buses-only--g1
- regulatory--end-of-maximum-speed-limit-30--g2
- regulatory--end-of-maximum-speed-limit-70--g1
- regulatory--end-of-maximum-speed-limit-70--g2
- regulatory--end-of-no-parking--g1
- regulatory--end-of-priority-road--g1
- regulatory--end-of-prohibition--g1
- regulatory--end-of-speed-limit-zone--g1
- regulatory--give-way-to-oncoming-traffic--g1
- regulatory--go-straight--g1
- regulatory--go-straight--g3

Assignment

- Adjust your concept hierarchy to the classes of the MTDS dataset
 - add missing subclasses (owl:subClassOf)
 - identify class equivalences (owl:equivalentClass)
- Do this with the appropriate prompt for LLM-based ontology development
- Generate additional competency questions that take into account this particular dataset

Remember to Keep a Record of...

- For each engineering step:
 - LLMs used
 - prompt (or series of prompts)
 - revised output file
- This information has to be provided in the final submission